

**10.6 Assignment 10: Explainable Artificial Intelligence (XAI): Concepts,
Taxonomies, Opportunities, and Challenges Toward Responsible AI**

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Benefits of Transparency In Machine Learning and Artificial Intelligence

Transparency and explainability in machine learning and artificial intelligence systems can provide benefits to both the users and populations affected by outcomes of these systems. First and foremost, public trust of these systems can increase their adoption and utilization. Machine learning and artificial intelligence already support human organizations and influence society in a multitude of ways. As AI becomes more available to the general public, fear of the unknown workings of these systems can unnecessarily impede their acceptance and advancement.

Transparency for users can assist in improving accuracy and functionality of models (Barredo Arrieta et al., 2020). When models are understood and measured, there is a better opportunity to identify error and bias. Increasing accuracy can save time and resources that would be wasted on error.

Fairness in Machine Learning and Artificial Intelligence

Barredo Arrieta et al. (2020), state a “fair” AI system will seek to consider: individual fairness, group fairness, and counterfactual fairness. Data used in models should be monitored for errors, work to eliminate harmful bias, utilize effective sample sizes, and optimal feature inclusion (Barredo Arrieta et al., 2020). One example of a machine learning system where “fairness” has a great impact on human life is in organ allocation. When using a model to decide who is granted an organ for donation and who is not, it is imperative that we strive to achieve the most fair model operation possible, as this directly affects human life. When considering patient demographics, medical histories, and wait times we want a model to have relevant features and appropriate data for ethical training and predictions.

My Perception of Bias, Fairness, Accountability, and Transparency

I see the concepts of bias, fairness, accountability, and transparency as critical to the future development of AI and our success in using these tools for good within humanity. We aim to achieve assistance, not loss, from these tools. The history of humankind is riddled with bias and unfair data, but machine learning and AI allow us to combine high levels of information that have the potential to smooth and rectify these challenges if we are intentional in doing so. There is a risk to perpetuate these

narratives if we don't utilize AI as a tool to lead us in an appropriately informed direction. Accountability and transparency support each other as we want people using these tools to be accountable through transparency. Transparency limits the power of users by demonstrating how decisions are made. The more transparent models are, the less they can be used for unethical gains.

Can Biased be Fair?

Returning to the example of organ allocation, there are instances of using machine learning in this realm where we can conclude bias is fair. For example, if a patient's health history indicates an organ transplant would not be successful, it would be fair to assign the organ to a different case where success could be achieved. The model is "discriminating" against those who might "waste" the resource. There have to be ethical parameters and definitions of success for the model to define this intended bias.

Another example could be wait time. If someone has been waiting longer or their situation is more urgent they might receive bias in the decision making process. What is "fair" in this process also has to be defined for the model. Is it first come first serve selection? Or if a patient's health is stable enough to wait while another's situation is urgent, who would get priority? These are the challenging situations where bias is necessary and can be considered fair to maximize saving human lives.

Conclusion

Machine Learning and Artificial Intelligence already influence many facets of society. As we continue to grow the use of these tools we need to do so intentionally and consciously to protect human interests. Efforts to understand these tools and their outputs help us build a better future with improved gain using Artificial Intelligence.

References

Barredo Arrieta, A., Díaz-Rodríguez, N., Del Ser, J., Bennetot, A., Tabik, S., Barbado, A., García, S., Gil-López, S., Molina, D., Benjamins, R., Chatila, R., & Herrera, F. (2020). *Explainable Artificial Intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI*. *Information Fusion*, 58, 82-115.
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